MILITARY SPECIFICATION

# THEOPHYLLINE AND GUAIFENESIN ELIXIR

This specification is approved for use by the Defense Personnel Support Center, Defense Supply Agency and is available for use by all Department and Agencies of the Department of Defense.

## 1. SCOPE

1.1 Scope. This specification covers Theophylline and Guaifenesin Elixir (See 6.2).

## 2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-C-186 - Containers, Packaging and Packing for Drugs, Chemicals and Pharmaceuticals. (see 6.3)

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Hq, Defense Personnel Support Center, ATTN: Directorate of Medical Materiel, DPSC-ATT, 2800 South 20th Street, Philadelphia, PA 19101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

FSC 6505

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposals shall apply.

U. S. DEPARTMENT OF HEALTH AND HUMAN SERVICES
FOOD AND DRUG ADMINISTRATION

Federal Food, Drug, and Cosmetic Act and Regulations Promulgated Thereunder.

(Application for copies should be addressed to the Food and Drug Administration, Washington, DC 20204.)

U. S. PHARMACOPEIAL CONVENTION, INC.

The United States Pharmacopeia (USP).

(Application for copies should be addressed to the Mack Publishing Company, Easton, PA 18042.)

The National Formulary (NF).

(Application for copies should be addressed to the Mack Publishing Company, Easton, PA 18042.)

3. REQUIREMENTS

3.1 Material. Shall be an elixir containing in each 5 ml [illegible] of Anhydrous Theophylline, 30.0 mg of Guaifenesin and [illegible] alcohol, within the designated assay limits.

3.1.1 Additions. The finished preparation shall contain suitable preservative(s). The coloring agent(s), flavoring agent(s), and the preservative(s) in the product shall be in the amounts as approved by the FDA which will make the preparation attractive and will properly preserve the preparation.

3.2 Finished preparation.

3.2.1 Assay. The elixir shall assay to contain not less than 90.0 percent and not more than 110.0 percent of the required amount of anhydrous Theophylline, and not less than 86.7 percent and not more than 113.3 percent of the required amount of Guaifenesin, when determined as specified in 4.5.1 and 4.5.2.



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3.2.2 Alcohol. The elixir shall assay to contain not less than 14.0 percent and not more than 16.0 percent alcohol V/V (93.3 percent to 106.7 percent of the required amount) when determined as specified in 4.5.3.

3.2.3 Identification. The theophylline and guaifenesin extracted from the elixir shall comply with the identity test specified in 4.5.4.

3.3 Ingredients.

3.3.1 Theophylline, anhydrous and alcohol. The anhydrous theophylline and alcohol used in the manufacture of the elixir shall comply with the tests, standards, and requirements of the USP, including any supplements or revisions thereto.

3.3.2 Guaifenesin. The Guaifenesin used in the manufacture of the elixir shall comply with the tests, standards, and requirements of the NF, including any supplements or revisions thereto.

3.3.3 Ingredients. All ingredients used in the manufacture of the elixir shall be in accordance with the tests, standards, and requirements of the USP or NF, including any supplements or revisions thereto; or, if not included in either of these compendia, the ingredients shall be of the highest pharmaceutical grade.

3.4 Expiration dating period. The expiration dating period shall be not less than 36 months.

3.4.1 Delivery. Not more than 3 months of the expiration dating period shall have elapsed at the time of delivery to the Government.

3.5 Workmanship. The material and its containers shall be free from defects which detract from their appearance or may impair their serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Supplier responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.



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4.1.1 Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and, (i) as to any expiration dated item, for the expiration dating period specified by the contractor for such item, or such longer period as may be required by regulation of any federal agency; or (ii) as to non-dated items, for not less than 3 years after delivery of the item to the Government.

4.1.2 No company supplying any ingredient(s) to the contractor will be considered an acceptable facility for the performance of any inspection requirements specified herein.

4.2 Lot. For purposes of this specification, a lot, batch, or control is that single, uniform, and homogeneous quantity of finished elixir produced from one formulation, subjected to the same compounding and manufacturing operation, and filled into final containers.

4.3 Sampling. Sampling shall be conducted in accordance with MIL-STD-105.

TABLE I. Sampling.

| For visual examination | Inspection level | AQL (percent defective) | Unit of product |
|---|---|---|---|
| Major | S-2 | 1.0 | Filled immediate containers |
| Minor | II | 2.5 | Filled immediate containers |

4.3.1 For end item testing. Noncompliance with any of the following is cause for rejection.

TABLE II. Criteria for end item testing.

| Characteristic | Requirement paragraph | Inspection level 1/ | Minimum sample |
|---|---|---|---|
| Assay | 3.2.1 | S-2 | 5 ) |
| Alcohol | 3.2.2 | S-2 | 5 } composite |
| Identification | 3.2.3 | S-2 | 5 ) |

1/ Where possible, the sample shall be used for two or more tests.



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4.4 Examination. The finished elixir shall be examined to determine compliance with all requirements of this specification. Nonconformance will be permitted to the extent indicated in 4.3 and 4.3.1.

4.4.1 Classification of defects. Examination will be conducted in accordance with the following classification of defects.

TABLE III. Classification of defects.

| Categories | Defects 1/ |
|---|---|
| Major | |
| 101 | Elixir not free from foreign matter. |
| 102 | Elixir has crystal formation. |
| 103 | Elixir not homogeneous. |
| Minor | |
| 201 | Exterior of immediate container is not free from contents or foreign matter. |
| 202 | Label not free from defacing marks and tears. |

1/ Inspection is not restricted to the classified possible defects listed above.

4.5 Tests. Assays.

4.5.1 Assay for Guaifenesin.

Reagents.

Chloroform, A.R.
Hydrochloric Acid, Conc., 37 percent, A.R.
1 N Sodium Hydroxide Solution. Dissolve 4.1 grams Sodium Hydroxide (A.R.) in purified water, dilute to 100.0 ml with purified water and mix.
Guaifenesin NF Reference Standard Solution.
Weigh accurately about 120 mg Guaifenesin NF Reference Standard and transfer to a 200 ml volumetric flask. Dissolve in chloroform and dilute to 200.0 ml; mix thoroughly. Prepare fresh for each assay.

Procedure.

Pipet 4.0 ml of the sample into a 125 ml separatory funnel. Allow the pipet to drain for 10 to 15 minutes; add 15 ml of 1 N Sodium Hydroxide Solution and mix by swirling. Place a pledget of cotton in the stem of the separatory funnel. Extract with three 25 ml portions of chloroform and one 20 ml portion of chloroform. Drain each chloroform



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with all applicable requirements of PPP-C-186b (See h.1) and as accordance

extract through cotton into a 100 ml volumetric flask. Note: Immediately add 3 ml concentrated hydrochloric acid to the aqueous layer in the separatory funnel; mix by swirling and save this solution for the theophylline assay. To the 100 ml volumetric flask containing the four chloroform extracts, add chloroform to make 100.0 ml and mix thoroughly. Piper 15.0 ml of the chloroform dilution of the sample extracts into a 100 ml volumetric flask. Dilute to 100.0 ml with chloroform and mix thoroughly. Pipet 6.0 ml of Guaifenesin NF Reference Standard Solution into a 100 ml volumetric flask. Dilute to 100.0 ml with chloroform and mix thoroughly. Concomitantly determine the absorbances of the sample and standard solutions at the maxima at about 276 nm and 310 nm in a 1 cm cell with a suitable spectrophotometer using chloroform as the blank. Calculate the amount of guaifenesin as follows:

Calculation:

$$\text{mg Guaifenesin per 5 ml} \quad \frac{Au^{276} - Au^{310}}{As^{276} - As^{310}} \times \frac{\text{gram standard}}{\text{ml sample}} \times 1000$$

where Au is the absorbance of the sample at the designated wavelengths.
As is the absorbance of the standard at the designated wavelengths.

4.5.2 Assay for Theophylline.

Reagents
Ether, USP
Hydrochloric Acid, Conc, 37 percent, A.R.
0.1N Hydrochloric Acid Solution
1N Sodium Hydroxide Solution
Theophylline USP Reference Standard Solution
Weigh accurately about 100 mg of Theophylline USP Reference Standard and transfer to a 500 ml volumetric flask. Dissolve in purified water and dilute to 500.0 ml with purified water; mix thoroughly. Prepare fresh for each assay.

Procedure.

Use the acidified aqueous extract from the guaifenesin assay. Remove the cotton from the stem of the separatory funnel containing the aqueous layer and extract the solution with 50 ml of Ether, USP. Drain the aqueous (lower) phase into a second 125 ml separatory funnel containing 25 ml of ether. Shake the second funnel for about 30 seconds and allow the layers to separate. Drain the aqueous layer into a 200 ml volumetric flask. Drain the ether layer in the second separatory funnel into the first separatory funnel; rinse the second separatory funnel with three 3 ml portions of ether and add rinsings to the first separatory funnel. Extract with two 25 ml portions of 0.1N Hydrochloric Acid Solution. Allow the layers to separate and drain each aqueous layer into the same 200 ml volumetric flask. Dilute to 200.0 ml with purified water and mix thoroughly. Pipet 10.0 ml of this Solution into a 250 ml volumetric flask. Pipet 10.0 ml of Theophylline USP Reference Standard Solution into a second 250 ml volumetric flask. Concomitantly treat the sample and standard solutions as follows: Add 25 ml 1 N Sodium Hydroxide Solution and dilute to 250.0 ml



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with purified water. Mix and immediately determine the absorbances at the maxima at about 274 nm and 320 nm in 1 cm cells with a suitable spectrophotometer using purified water as a blank. Calculate the amount of anhydrous theophylline as follows:

Calculation:

$$\text{mg anhydrous theophylline per 5 ml} = \frac{Au^{274} - Au^{320}}{As^{274} - As^{320}} \times \frac{\text{gram standard}}{\text{ml sample}} \times 2000$$

where Au is the absorbance of the sample at the designated wavelengths.
As is the absorbance of the standard at the designated wavelengths.

4.5.3 Assay for Alcohol. Shall be determined by either the USP distillation method or the USP gas-liquid chromatographic method for the determination of alcohol.

4.5.4 Identity. Place 10 ml sample in a 200 ml erlenmeyer flask. Add 10 ml purified water, 5 ml 0.1N Hydrochloric Acid and 2 grams chromatographic grade aluminum oxide. Shake vigorously for about 20 seconds. Allow to stand 40 to 60 seconds and again shake as before. Repeat shaking and standing 3 to 5 times and/or until the solution has become considerably lighter and the aluminum oxide is dark colored. Filter through Whatman No. 1 or equivalent filter paper, collecting the filtrate in a second erlenmeyer flask. Add 20 ml 1 percent mercuric acetate reagent solution (dissolve 1 gram Mercuric Acetate A.R. in purified water containing 1 ml glacial acetic acid and add purified water to make 100 ml). Shake and let stand as before until mercuric theophyllinate has precipitated. Filter through Whatman No. 1 or equivalent filter paper into a 125 ml separatory funnel (save the filtrate for the identification of guaifenesin). Wash the precipitate thoroughly with several portions of purified water; discard washings and dry precipitate in 100° C oven for about 30 minutes. Save this precipitate for the identification of theophylline. To the filtrate containing guaifenesin add 2 ml 10 percent Sodium Hydroxide, mix and check pH with litmus paper. If necessary, add additional 10 percent Sodium Hydroxide in 1 ml volumes until litmus turns blue and add an additional 1 ml 10 percent Sodium Hydroxide. Extract with two 25 ml portions of chloroform, collect the chloroform extracts in a 100 ml beaker. Evaporate the chloroform extract to dryness on a steam bath. Scrape the residue until it dries completely; this residue is guaifenesin. The theophylline and the guaifenesin shall each have an infrared absorption spectrum curve qualitatively comparable to that of a similar preparation of USP Theophylline Reference Standard and NF Guaifenesin Reference Standard. Record the infrared spectrum from 2 to 16 microns using air as a reference.



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4.5.1 Assay ... shall contain not less than 98.0 percent and not more than 115.0

4.6 Packaging inspection. The inspection of the intermediate package, packing and marking for shipment and storage shall be in accordance with quality assurance provisions of the applicable container specification and the marking requirements of MIL-STD-129.

4.7 Labeling inspection. Sampling shall be in accordance with MIL-STD-105 and the unit of product shall be the unit of issue. The inspection lot shall be the number of units of issue of each batch, lot, control number, etc., offered for inspection at one time. The inspection level for labeling shall be S-2 and the acceptance number shall be zero.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with all applicable requirements of PPP-C-186 (See 6.3) and as specified herein.

5.2 Packaging. Packaging shall be level A or C, as specified (See 6.1).

5.2.1 Unit of issue. One bottle containing one gallon (3.78 liters) as specified, constitutes one unit of issue.

5.2.2 Level A.

5.2.2.1 Immediate container. The immediate container shall be a commercially available bottle considered acceptable by the Food and Drug Administration (FDA) for the product contained therein and shall be of such compatibility that neither the contents nor the container are altered in any way by each other.

5.2.2.2 Unit package. Each unit shall be packaged as specified in 5.2.10 of PPP-C-186.

5.2.3 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.3 Packing. Packing shall be level A, B or C, as specified (see 6.1).

5.3.1 Level A. Four (4) units of issue shall be packed as specified in 5.4.4 of PPP-C-186.

5.3.2 Level B. Four (4) units of issue shall be packed as specified in 5.4.3.4 of PPP-C-186.



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5.3.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.4 Marking.

5.4.1 Labeling. Labeling shall be in accordance with the requirements of the Federal Food, Drug and Cosmetic Act and shall include the information required below:

5.4.1.1 Immediate container. Each immediate container label shall bear the following information:

(a) label information in accordance with commercial practice.
(b) the expiration date.

5.4.1.2 Unit package. Each unit package label shall bear the following information:

(a) label information in accordance with commercial practice.
(b) the National Stock Number.

NOTES: (1) The National Stock Number shall not be preceded by any prefix.
(2) Strip labeling may be used.
(3) Label adhesion test shall not apply.
(4) If a unit package is not supplied, the National Stock Number shall appear on the immediate container label.

5.4.2 Exterior container. Exterior container shall be marked as specified in 5.5.4 of PPP-C-186. Type I shelf life marking as specified in MIL-STD-129 shall also be shown. Marking shall include the legend: "DO NOT PERMIT TO FREEZE."

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) National Stock Number.
(c) Selection of applicable levels of packaging and packing (See 5.2 and 5.3).

6.2 This specification covers the following item(s):

| National Stock Number | Item Identification |
|---|---|
| 6505-01-026-0350 | THEOPHYLLINE AND GUAIFENESIN ELIXIR, 1 Gal (3.78 liters) |



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6.3 All references to PPP-C-186 refere to Interim Federal Specification PPP-C-00186B which is mandatory for use until superseded by a revision.

Preparing activity:
Defense Supply Agency - DM

Project No. 6505-1819



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